Maruf Adesina Lawal

Curriculum Vitæ

Department of Mathematics
University of Tennessee
Knoxville TN
37916, USA
+1 865-910-4535
mlawal@vols.utk.edu

Research Interests

- Mathematical Biology and Ecology, Infectious Diseases Modeling, Natural Resources Modeling, Optimal Control Theory, Machine Learning/AI for Health, Data-Driven Modeling of Biological Systems, Data Analysis, Computational Mathematics.

Education

Dec. 2026 (Expected) **PhD, Mathematics, *Department of Mathematics***, *University of Tennessee (UTK)*, Knoxville, USA
Dissertation (in progress): *Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy and An Optimal Control Framework for Multi-Species Harvesting in the Southern Black Sea*
Advisor: Prof. Suzanne Lenhart

May. 2023 **Master of Science, Mathematics, *Department of Mathematics***, *University of Tennessee (UTK)*, Knoxville, USA

Feb. 2015 **B.Tech, Mathematics, *Department of Pure and Applied Mathematics***, *Ladoke Akintola University of Technology (LAUTECH)*, Ogbomoso, Nigeria
Distinction : *first class honours*
Thesis : *Mathematical modeling of a standard incidence epidemiological model with relapse*
Advisor: Dr. Samson Olaniyi

Professional Experience

2022 - Current **Graduate Teaching Associate**, *University of Tennessee (UTK)*, Knoxville, Tennessee, USA

2021 - 2022 **Graduate Teaching Assistant**, *University of Tennessee (UTK)*, Knoxville, Tennessee, USA

2019 - 2021 **High School Mathematics Teacher**, *Learn-With-Ease (LWE) Schools*, Lagos, Nigeria

2018 - 2021 **Tutor**, *Tractrain Consulting*, Lagos, Nigeria

2015 - 2017 **Teaching Assistant**, *Federal Polytechnic Kaura Namoda (FPKN)*, Zamfara, Nigeria

Publications

M. A. Lawal, M. Demir, M. R. Kelly, and S. Lenhart, An Optimal Control Framework for Multi-Species Harvesting in the Southern Black Sea. *Natural Resource Modeling* **39**:e70044, 2026. doi:10.1111/nrm.70044.

C. W. Chukwu, Maruf A. Lawal, D. Aldila, and G. Obaido, Evaluating Syphilis Transmission Among MSM Population: A Mathematical Model for Reinfection, Two-Stage Treatment and Treatment Failure. *Front. Public Health* **14**:1841745, 2026. doi:10.3389/fpubh.2026.1841745.

C. W. Chukwu, M. A. Lawal, K. R. Cheneke, and D. Wanduku, Simulation-Based Predictive Modeling of Behavioral Heterogeneity in Chlamydia Transmission: Insights from U.S. Surveillance Data (*Under Review*).

Samson Olaniyi, Maruf A. Lawal, and Olawale S. Obabiyi, Stability and Sensitivity Analysis of a Deterministic Epidemiological Model with Pseudo-recovery. *IAENG International Journal of Applied Mathematics*, **46**(2):160–167, 2016.

M. A. Lawal, C. Edholm, H. Correia, F. Agusto, S. Lenhart, W. Chukwu, D. Goldsman, F. Chirove, J. White, and S. Dobson, Modeling the Interplay of Vaccine Hesitancy and Disease Dynamics (*Manuscript in preparation*).

Invited Talks

Jan. 2026 **Optimal Sustainable Management of Multi-Species Harvesting in the Southern Black Sea Ecosystem**, *2026 Joint Mathematics Meetings (JMM)*, Washington, DC, USA

March. 2025 **Optimal Sustainable Management of the Horse Mackerel Harvesting in the Southern Black Sea Coast**, *2025 SIAM Southeastern Atlantic Section Meeting*, Knoxville, TN, USA

Jan. 2025 **Optimal Sustainable Management of the Horse Mackerel Harvesting in the Southern Black Sea Coast**, *2025 Joint Mathematics Meetings (JMM)*, Seattle, WA, USA

March 2024 **Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy**, *103rd Annual MAA Southeastern Sectional Meeting*, Knoxville, TN, USA

Jan. 2024 **Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy**, *2024 Joint Mathematics Meetings (JMM))*, San Francisco, CA, USA

Contributed Talks

June 2026 **Modeling the Interplay of Vaccine Hesitancy and Disease Dynamics**, *Modeling the Invisible Workshop, Georgia State University*, Atlanta, GA, USA

Nov. 2024 **Optimal Sustainable Management of the Horse Mackerel Harvesting in the Southern Black Sea Coast**, *Seminar: Mathematics Presentations, University of Tennessee*, Knoxville, TN, USA

Sept. 2024 **Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy**, *Seminar: Mathematics Presentations, University of Tennessee*, Knoxville, TN, USA

Conferences and Workshops Attended

June 2026 **Modeling the Invisible Workshop**, *Atlanta, GA, USA*

Jan. 2026 **2026 Joint Mathematics Meetings (JMM)**, *Washington, DC, USA*

Nov. 2025 **Masamu Advanced Study Institute (MASI) Convening Research Workshop (Virtual)**

Nov. 2025 **2025 Virtual SAMSA Masamu Advanced Study Institute (MASI) Convening**

June/July 2025 **2025 Institute for Mathematics and its Applications (IMA) Math-to-Industry Boot Camp**, *virtual*
Project: Proactive Housing Support: using public data to predict and prevent evictions

June. 2025 **2025 Institute for Mathematical and Statistical Innovation (IMSI) Data Science Boot Camp**, *virtual*

April. 2025 **2025 One Health Student Leadership Summit**, *University of Tennessee, Knoxville, TN, USA*

March. 2025 **2025 SIAM Southeastern Atlantic Section Meeting**, *University of Tennessee, Knoxville, TN, USA*

Jan. 2025 **2025 Joint Mathematics Meetings (JMM)**, *Seattle, WA, USA*

March 2024 **103rd Annual Mathematical Association of America (MAA) Southeastern Sectional Meeting**, *University of Tennessee, Knoxville, TN, USA*

Jan. 2024 **2024 Joint Mathematics Meetings (JMM)**, *San Francisco, CA, USA*

July 2023 **Society for Mathematical Biology Annual Meeting**, *Ohio State University*, *Ohio, OH, USA*

June 2022 **Mathematical Problems in Industry (MPI) Workshop**, *Worcester Polytechnic Institute*, *MA, USA*
Project: Patient data generation and machine learning utilizing clinical data to predict the severity of heart failure

June 2022 **Graduate Student Mathematical Modeling Camp (GSMMC)**, *University of Delaware*, *DE, USA*
Project: Asymmetric division and modeling of interacting cell populations in the colonic crypt

Nov. 2022 **Twelfth Annual Masamu Advanced Study Institute (MASI) and Research Workshop**, *as part of the 2021 Southern African Mathematical Sciences Association (SAMSA) Conference held in Maputo, Mozambique (Virtual)*, *Online*

Nov. 2021 **Eleventh Annual Masamu Advanced Study Institute (MASI) and Research Workshop**, *as part of the 2021 Southern African Mathematical Sciences Association (SAMSA) Conference*, *Online*

Courses Taught at University Level

Instructor of Record

- Statistical Reasoning (MATH 115)–UTK — Spring, Fall 2026
- Basic Calculus (MATH 125)–UTK — Spring 2025
- Basic Calculus (MATH 125)–UTK — Spring, Fall 2024
- Basic Calculus (MATH 125)–UTK — Spring, Fall 2023
- Basic Calculus (MATH 125)–UTK — Fall 2022
- Trigonometry and Analytic Geometry (MATH 122) – FPKN — Semester 2, 2016

Co-Instructor

- Calculus I (MATH 141) – UTK — Fall 2025
- Calculus I-B Infused with Precalculus (MATH 132) – UTK — Fall 2025

Teaching Assistant

- Basic Calculus (MATH 125)–UTK — Spring 2022
- College Algebra (MATH 119) – UTK — Fall 2021

Awards and Fellowships

- University of Tennessee Open Publishing Support Fund (OPSF) Award ($1800) — Sept 2026
- Winner, "Modeling the Invisible: Competition on Forecasting Viral Spread with Limited Data," Modeling the Invisible Workshop — June 2026
- FY23 Graduate School Fellowship ($1018) — Jan 2026
- University of Tennessee Graduate Student Senate (GSS) Travel Award ($800) — Jan. 2026
- FY23 Graduate School Fellowship ($1018) — Aug 2025
- FY22 University of Tennessee Graduate School Fellowship ($2000) — 2025
- AMS BEGIN Bootcamp Travel Grant ($300) — Jan. 2025
- 2025 AMS Joint Mathematics Meetings (JMM) Graduate Student Travel Grant ($1430) — Jan. 2025
- University of Tennessee Graduate Student Senate (GSS) Travel Award ($500) — Jan. 2025
- University of Tennessee Math Department Summer Dissertation Fellowship ($4000) — June 2024
- FY21 University of Tennessee Graduate School Fellowship ($1249) — 2024
- FY22 University of Tennessee Graduate School Fellowship ($2000) — 2024
- University of Tennessee Math Department Award ($2300) — 2024
- University of Tennessee Graduate Student Senate (GSS) Travel Award ($650) — Jan. 2024
- University of Tennessee Math Department Award ($2000) — 2023
- FY22 University of Tennessee Graduate School Fellowship ($2000) — 2023
- Yueh-er, Hong-hsu and Clarence Cheng Kuo Fellowship, at the 2023 UTK Mathematics Honors Day ($1000) — April 2023
- University of Tennessee Math Department Academic Excellence Fellowship ($1660) — 2022
- University of Tennessee Graduate School Award ($400) — 2022
- FY22 University of Tennessee Graduate School Fellowship ($2000) — 2022
- 2015 National Mathematics Competition for University Students (NAMCUS) Gold Medal Winner held at the National Mathematical Center, Abuja, Nigeria — April 2015

Service

Jan. 2025 **Reviewer** , *Undergraduate Poster Session, Joint Mathematics Meetings (JMM)*, Seattle, WA, USA

2024 - 2025 **Graduate Student Representative**, *University of Tennessee Technology Fee Advisory Board*

2024 - 2025 **Elected Senator**, *University of Tennessee Graduate Student Senate (GSS)*

Jan. 2024 **Reviewer** , *Undergraduate Poster Session, Joint Mathematics Meetings (JMM)*, San Francisco, CA, USA

Oct. 2023 **Panelist** , *University of Tennessee SIAM Graduate Student Internship/Workshop Experiences Panel*

Computing Proficiency

Programming **Python, MATLAB, R, Maple, LaTeX**

Applications **Microsoft office Package (Word, Excel, Powerpoint)**

Journal Review Work

- Journal of Biological Dynamics
- Scientific Reports (SR) Springer Nature
- PLOS One
- Mathematical Modelling and Applications
- Mathematical Modelling and Numerical Simulation with Applications

Professional Affiliations

- American Mathematical Society (AMS)
- Society for Mathematical Biology (SMB)
- Mathematical Association of America (MAA)
- Society for Industrial and Applied Mathematics (SIAM)
- Models of Infectious Disease Agent Study (MIDAS)
- Nigerian Mathematical Society (NMS)

Online Courses/Certifications

- Complete Guide to R: Wrangling, Visualizing, and Modeling Data - *LinkedIn Learning* Oct, 2024
- Machine Learning with MATLAB - *MathWorks Training Services* May 2024
- Hands-On Advanced Python: Data Exploration and Manipulation - *LinkedIn Learning* May 2024
- Python for Data Science and Machine Learning Essential Training - *LinkedIn Learning* May 2024
- UTK Department of Mathematics GTA Teaching Advising Program May 2023